

M03 Review / Repair - Teaching RC2

OPEN ONLY AFTER A GENUINE SAVED ATTEMPT

How to use this

Do not read every answer. Find the competency you missed, compare the reasoning pattern, repair only that weakness, close this review, then perform a fresh retry.

P01 repair note

If the business says keep every eligible left-side order, start from that eligible order set and use LEFT JOIN. Be careful: a WHERE filter on c.* after the join can remove NULL matches.

STOP - check your understanding

Close this review. Can you now solve a changed version without copying this wording?

Do not turn the page until you can answer in your own words.

P02 repair note

The key evidence is 30 base orders -> 35 joined rows while `COUNT(DISTINCT order_id)=30`. The relationship, not corruption, caused repeated order rows.

STOP - check your understanding

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P03 repair note

UNION ALL is correct when every source event should survive. Validate 27 Completed orders + 10 tickets = 37 events.

STOP - check your understanding

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P04 repair note

Keep raw and standardized values side by side. SELECT expressions do not update source values.

STOP - check your understanding

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P05 repair note

State the metric definition before calculating: $\text{return_date} - \text{order_date}$. Max delay in supplied data is 4 days.

STOP - check your understanding

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P06 repair note

Run the Completed-order average inner query first (~1592.02 net_sales), then compare outer Completed rows. Expected 13 rows.

STOP - check your understanding

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P07 repair note

Good CTE names describe meaning and grain. order_sales = one row/order; completed_avg = one row total.

STOP - check your understanding

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P08 repair note

A window average should preserve 27 Completed rows. Validate regional averages with a separate GROUP BY.

STOP - check your understanding

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P09 repair note

Latest row pattern: ROW_NUMBER() OVER(PARTITION BY entity ORDER BY timestamp DESC, stable_tiebreaker DESC), then rn=1.

STOP - check your understanding

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P10 repair note

LAG must partition by the entity whose sequence you want. First event per entity has NULL previous value.

STOP - check your understanding

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P11 repair note

All-order return rate: East $0/8$, North $2/7$, South $0/7$, West $3/8$. Numerator and denominator are both distinct orders.

STOP - check your understanding

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P12 repair note

PostgreSQL schema.table explicitly identifies an object. PK protects identity; NOT NULL required fields; CHECK domain rule.

STOP - check your understanding

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P13 repair note

Index claims need evidence. Small tables may keep Seq Scan. Compare representative EXPLAIN/ANALYZE evidence, not belief.

STOP - check your understanding

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P14 repair note

A useful view contract includes name, grain, business filter, columns, sources, owner/change expectations and validation.

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